

Embodiment × Minimalism

UBISS 2026 Workshop A

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Outline

- ▶ What is embodiment?
- ▶ How does embodiment relate to minimalism in robotics?
- ▶ A light smorgasbord of (hopefully inspiring) examples

Embodiment

Embodiment is a state of having a physical body.

An *embodied agent* has meaningful sensory-motor interactions with the environment.

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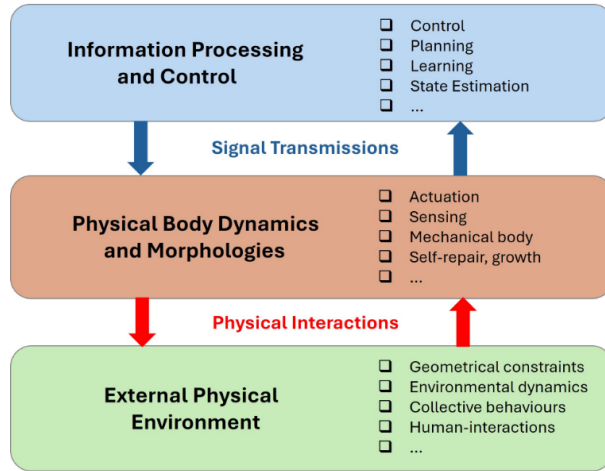
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- ▶ Understanding embodied intelligence helps us shift information processing from the “brain” to the body

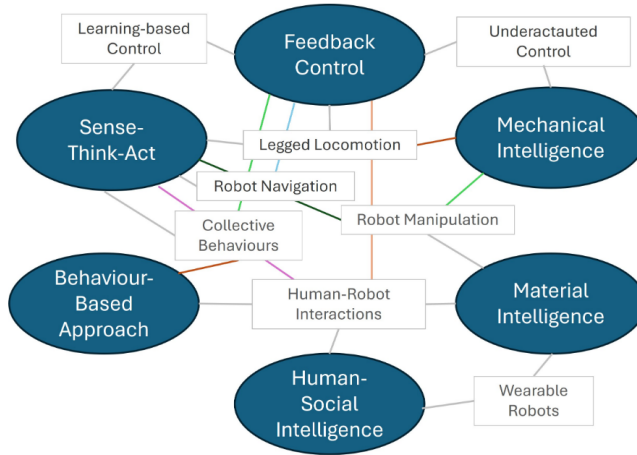
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- ▶ EI conference: <https://embodied-intelligence.org/>
- ▶ Understanding embodied intelligence helps us shift information processing from the “brain” to the body
- ▶ Reduces explicit computational requirements by offloading

Embodied Intelligence



Embodied Control



Embodiment and Minimalism in Robotics

Geometry

- ▶ visibility
- ▶ compliance
- ▶ manipulation

Geometry

- ▶ part orientation without sensing
- ▶ Bouncing robots (external video)

Dynamics

- ▶ funnel chaining
- ▶ oscillations and fixed points
- ▶ Passive Dynamic Walkers

Expressivity

- ▶ lavers paper
- ▶ keepon spoon dancer

Human Response to Embodied Agents

- ▶ you know what it is

More Examples

Proprioception vs. Exteroception

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- ▶ Exteroception: sensing the external world

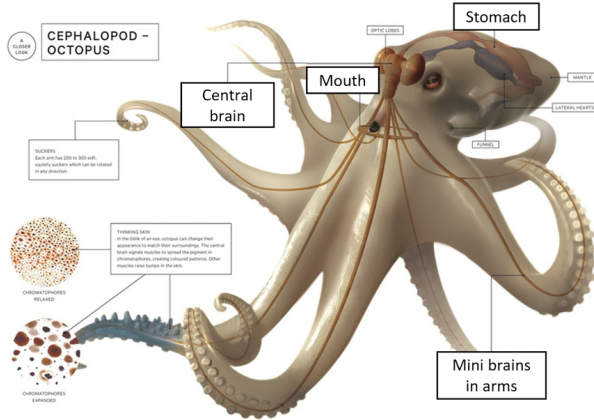
Proprioception vs. Exteroception

- ▶ Proprioception: sensing of “internal” states
- ▶ Exteroception: sensing the external world
- ▶ Let's try it!

Embodied Wallfollowing

► Mouse

Distributed Information Processing



Soft Robots

- ▶ passive fishes
- ▶ foambots

Collective Embodied Intelligence

- ▶ brazil nut
- ▶ boids
- ▶ collective braitenberg
- ▶ swarmulators

A Collective Embodiment Exercise

Conclusion

There is so much

- ▶ Could have whole week workshop on this topic

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- ▶ Instead, discussion: connections to docking exercise?

A Useful Constraint for Design

Source: Eva & Franco Mattes

Source: gatsby.bot

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